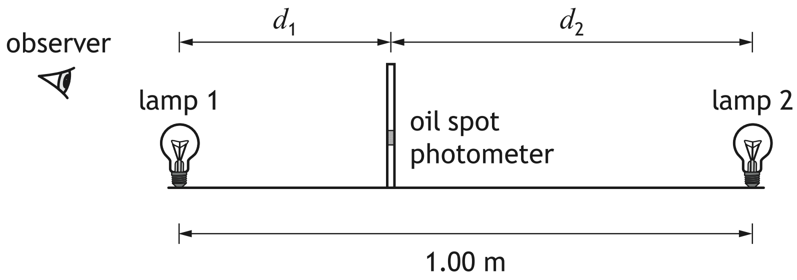
DO NOT WRITE IN THIS MARGIN

8. A group of students is conducting an experiment to investigate the relationship between the power output of two small filament lamps and the relative brightness of each lamp.

The relative brightness can be determined using an oil spot photometer.

The experiment is set up as shown.



The oil spot photometer is a piece of card with an oil spot on it, which is used to compare the brightness of two light sources on either side of the photometer. When the relative brightness of the light on either side of the photometer is equal, no spot is observed.

* X 8 5 7 7 6 0 1 2 6 *

8. (continued)

MARKS DO NOT WRITE IN THIS MARGIN

(a) The students keep the brightness of lamp 1 constant.

The brightness L_2 of lamp 2 is measured using a light meter.

The photometer is moved until no spot is observed.

The distance d_2 between lamp 2 and the photometer is measured.

The brightness of lamp 2 is then increased and the process is repeated.

The following results are obtained.

L_2 (units)	d_2 (m)
5.00	0.350
10.00	0.495
15.00	0.606
20.00	0.700

Use **all** of the data in the table to establish that the relationship between the brightness L_2 and the distance d_2 when no oil spot is observed is

$$\frac{L_2}{d_2^2} = \text{constant}$$ 3

Space for working and answer

(b) Suggest why small, spherical lamps were used for this experimental procedure. 1

X857760127

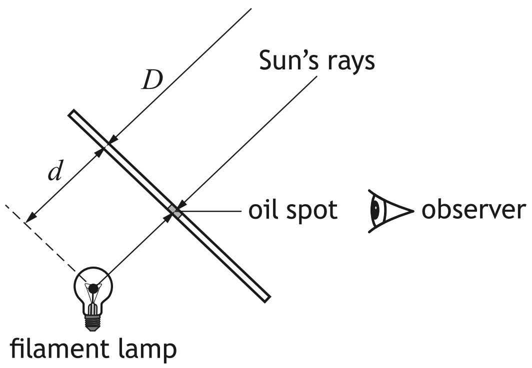
[Turn over

8. (continued)

(c) The students repeat the experiment outside, to determine the luminosity of the Sun.

Luminosity is a measure of radiated electromagnetic power, measured in watts.

The set up for this experiment is shown below.



The luminosity of the Sun can be determined using the relationship

$$\frac{L_{lamp}}{d^2} = \frac{L_{Sun}}{D^2}$$

where: L_{lamp} is the luminosity of the lamp, in watts

L_{Sun} is the luminosity of the Sun, in watts

d is the distance between the lamp and the photometer, in metres

D is the approximate distance from the Sun to Earth, in metres.

The luminosity of the lamp is 1.0×10^2 W.

The approximate distance between the Sun and the Earth is 1.5×10^{11} m.

The students adjust the photometer position until the oil spot disappears and note the distance between the centre of the lamp and photometer.

The students repeat the process five times and record the following results:

d = 0.88 m, 0.86 m, 0.90 m, 0.89 m, 0.86 m

MARKS DO NOT WRITE IN THIS MARGIN

8. (c) (continued)

(i) Determine the mean value for d. **1**

Space for working and answer

(ii) Calculate the random uncertainty in the mean value for d. **2**

Space for working and answer

(iii) Using the data from this experiment, determine the luminosity of the Sun.

An uncertainty in this value is not required. **2**

Space for working and answer

[Turn over